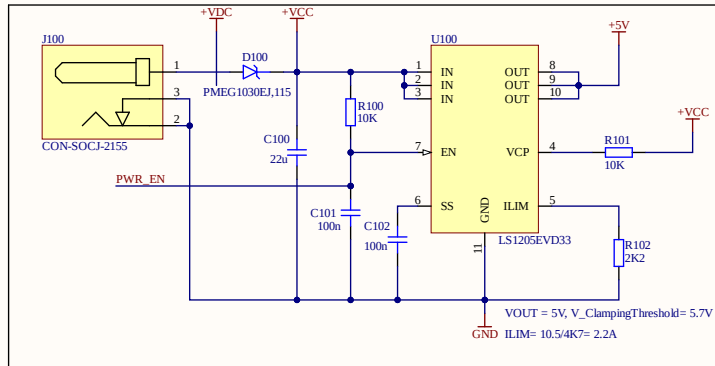
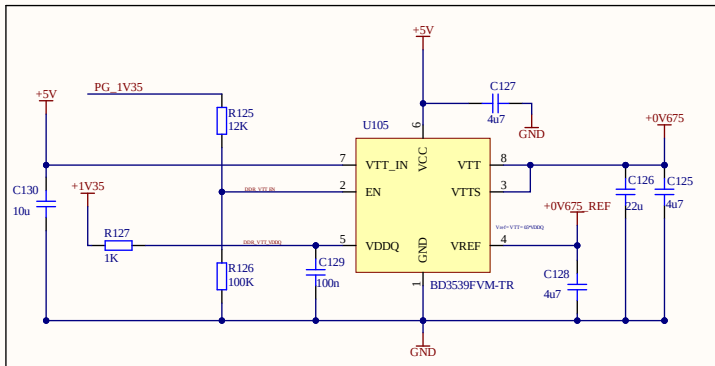


[1] POWER

Input Power & eFuse



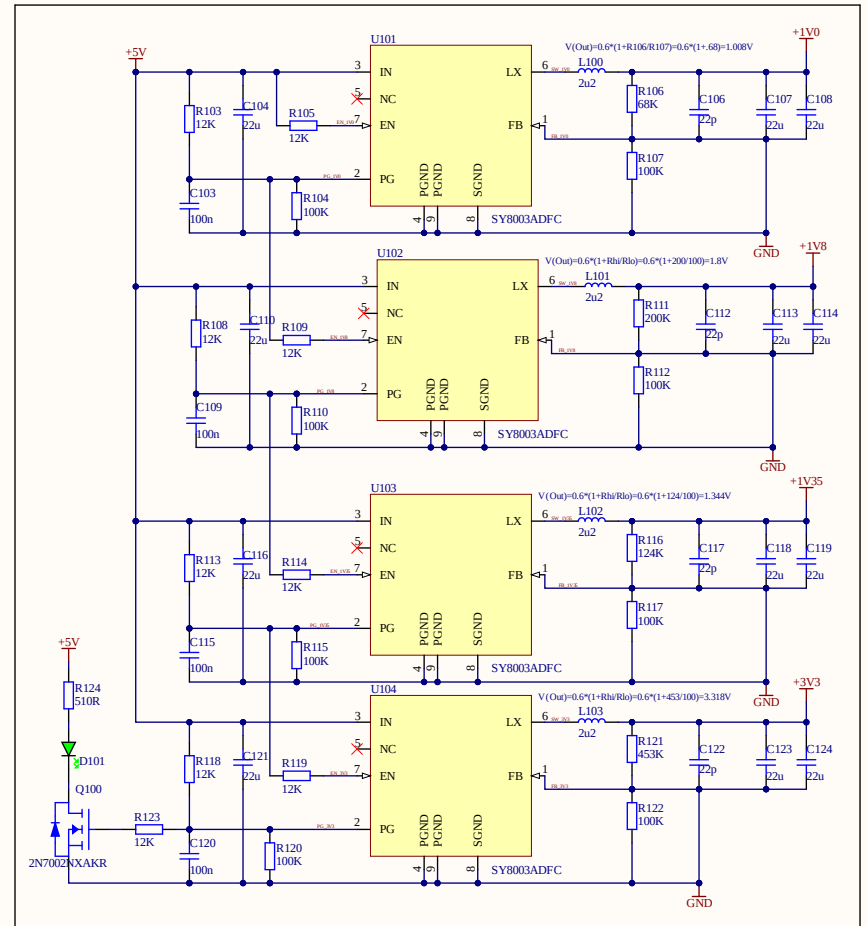
DDR3L VTT Regulator



Buck Converters

Power sequence: V1 -> V2 -> V3 -> V4

Power sequence: +1V0 -> +1V8 -> +1V35 -> +3V3



Title: Power			
Size: A3	Number: 1	Revision: A	
Date: 13-Nov-23	Time: 5:54:14 AM	Sheet 1 of 10	
File: C:\Users\Gebraiker\Downloads\Courses\Advanced Hardware Design (Phil Lab)\AltiumProject\ZettBrettBoard\Power.SchDoc			

[2] Zynq Power and Decoupling

Main Supplies (INT, AUX, DDR, MIO)

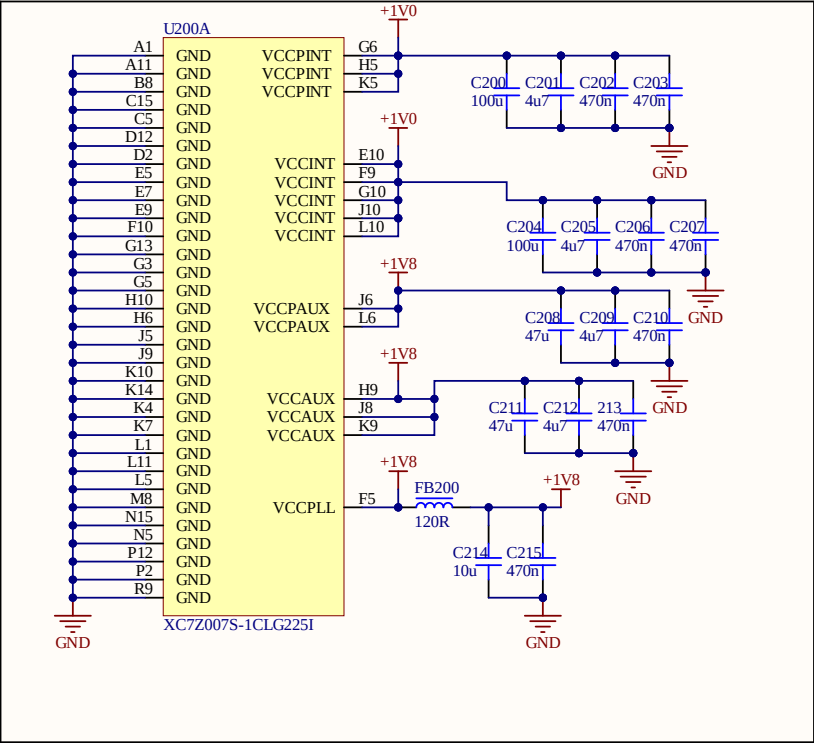


Table 3-1: Required PCB Capacitor Quantities per Device (PL)

Package	Device	VCCINT					VCCBRAM				VCCAUX			VCCAUX_IO			VCCO per Bank ⁽³⁾⁽⁴⁾			Bank 0
		680 μ F	330 μ F	100 μ F	4.7 μ F	0.47 μ F	100 μ F	47 μ F	4.7 μ F	0.47 μ F	47 μ F	4.7 μ F	0.47 μ F	47 μ F	4.7 μ F	0.47 μ F	47 μ F or 100 μ F	4.7 μ F	0.47 μ F	47 μ F
CLG225	Z-7007S	0	0	1	1	2	NA	NA	NA	NA	1	1	1	NA	NA	NA	1	2	4	1
CLG400	Z-7007S	0	0	1	1	2	0	1	1	1	1	1	1	NA	NA	NA	1	2	4	1
CLG225	Z-7010	0	0	1	1	2	NA	NA	NA	NA	1	1	1	NA	NA	NA	1	2	4	1
CLG400	Z-7010	0	0	1	1	2	0	1	1	1	1	1	1	NA	NA	NA	1	2	4	1

Table 3-2: Required PCB Capacitor Quantities per Device (PS)

Package	Device	VCCPINT			VCCPAUX ⁽¹⁾			VCCO_DDR			VCCO_MIO0			VCCO_MIO1			VCCPLL ⁽²⁾⁽³⁾
		100 μ F	4.7 μ F	0.47 μ F	100 μ F	4.7 μ F	0.47 μ F	100 μ F	4.7 μ F	0.47 μ F	100 μ F	4.7 μ F	0.47 μ F	100 μ F	4.7 μ F	0.47 μ F	
CLG225	Z-7007S	1	1	3	1	1	1	1	1	4	1	1	1	1	1	1	
CLG400	Z-7007S	1	1	3	1	1	1	1	1	4	1	1	1	1	1	1	
CLG225	Z-7010	1	1	3	1	1	1	1	1	4	1	1	1	1	1	1	
CLG400	Z-7010	1	1	3	1	1	1	1	1	4	1	1	1	1	1	1	
CLG485	Z-7012S	1	1	3	1	1	1	1	1	4	1	1	1	1	1	1	
CLG400	Z-7014S	1	1	3	1	1	1	1	1	4	1	1	1	1	1	1	
CLG484	Z-7014S	1	1	3	1	1	1	1	1	4	1	1	1	1	1	1	
CLG485	Z-7015	1	1	3	1	1	1	1	1	4	1	1	1	1	1	1	
CLG400	Z-7020	1	1	3	1	1	1	1	1	4	1	1	1	1	1	1	

Zynq Config, JTAG

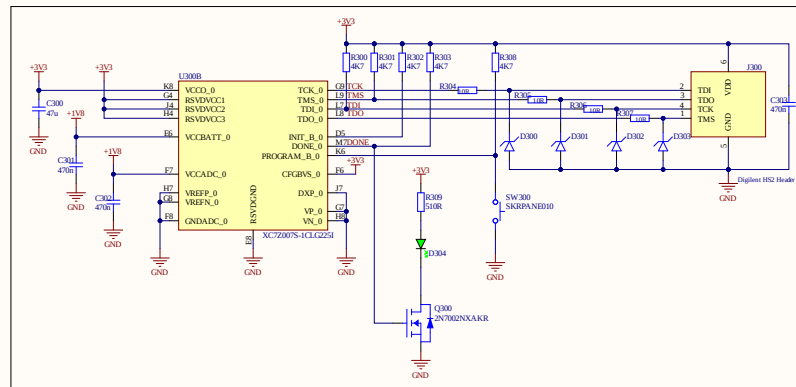


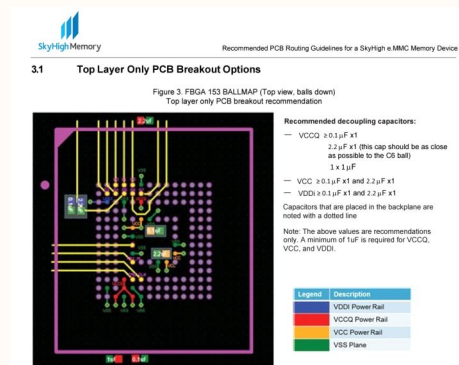
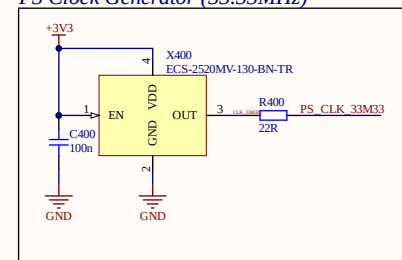
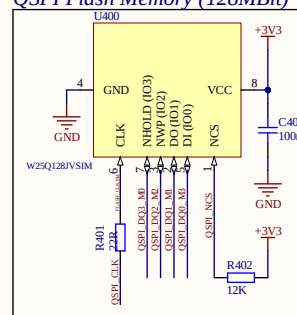
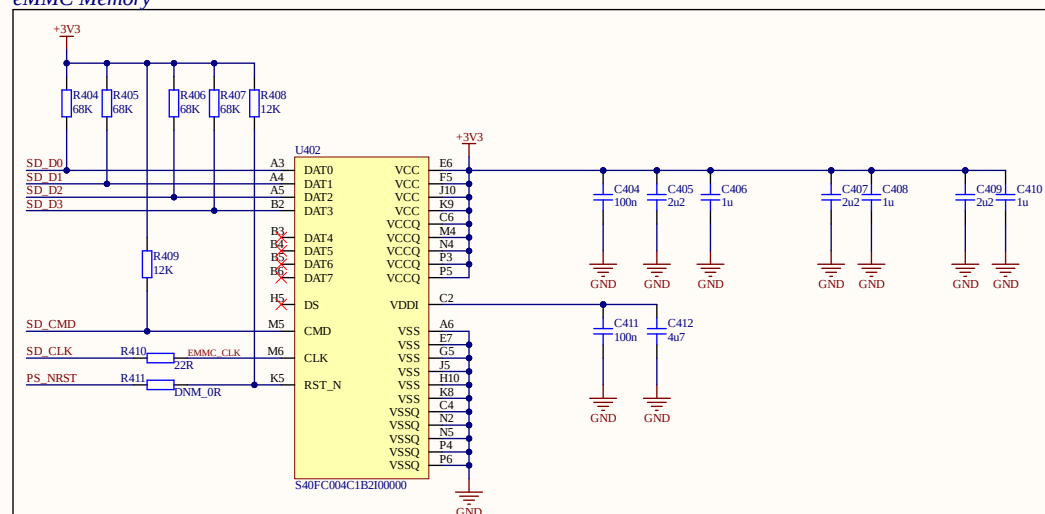
Table 6-4: Boot Mode MIO Strapping Pins

Pin-signal / Mode	MIO[8]	MIO[7]	MIO[6]	MIO[5]	MIO[4]	MIO[3]	MIO[2]
	VMODE[1]	VMODE[0]	BOOT_MODE[4]	BOOT_MODE[3]	BOOT_MODE[2]	BOOT_MODE[1]	BOOT_MODE[0]
Boot Devices							
JTAG Boot Mode; cascaded is most common ⁽¹⁾				0	0	0	JTAG Chain Routing 0: Cascade mode 1: Independent mode
NOR Boot ⁽²⁾				0	0	1	
NAND				0	1	0	
Quad-SPI ⁽³⁾				1	0	0	
SD Card				1	1	0	
Mode for all 3 PLLs							
PLL Enabled			0	Hardware waits for PLL to lock, then executes BootROM.			
PLL Bypassed			1	Allows for a wide PS_CLK frequency range.			
MIO Bank Voltage ⁽⁴⁾							
	Bank 1	Bank 0	Voltage Bank 0 includes MIO pins 0 thru 15.				
2.5 V, 3.3 V	0	0	Voltage Bank 1 includes MIO pins 16 thru 53.				
1.8 V	1	1					

1. JTAG cascaded mode is most common and is the assumed mode in all the references to JTAG mode except where noted.
2. For secure mode, JTAG is not enabled and MIO[2] is ignored.
3. The Quad-SPI and NOR boot modes support execute-in-place (this support is always non-secure)
4. Voltage Banks 0 and 1 must be set to the same value when an interface spans across these voltage banks. Examples include NOR, 16-bit NAND, and a wide TPIU test port. Other interface configuration may also span the two banks.

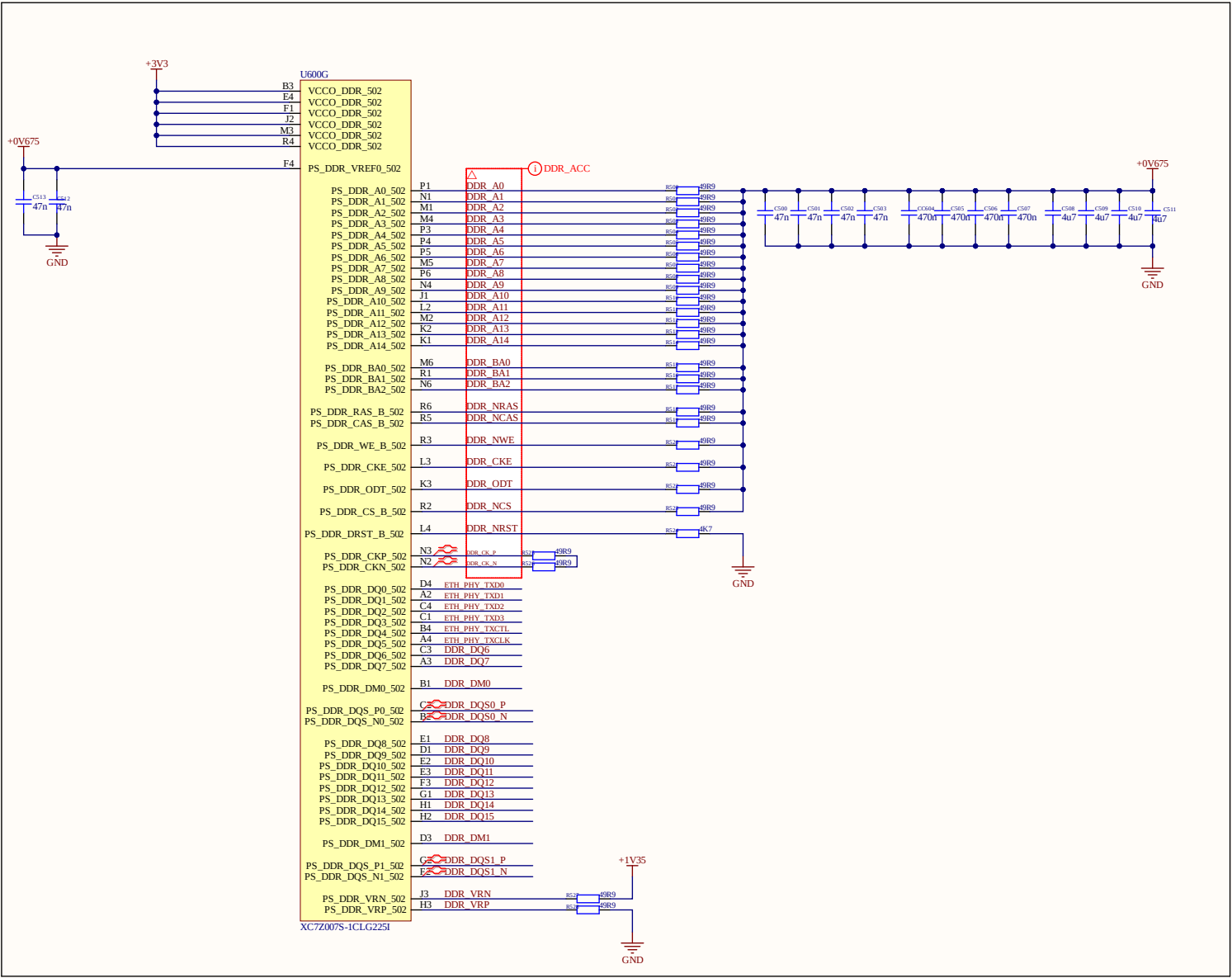
Title: Zynq (Config, JTAG, Debug)			
Size: A2	Number: 3	Revision: A	
Date: 13-Nov-23	Time: 5:54:15AM	Sheet 3 of 10	
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Zynq PS (Bank 500 & 501)



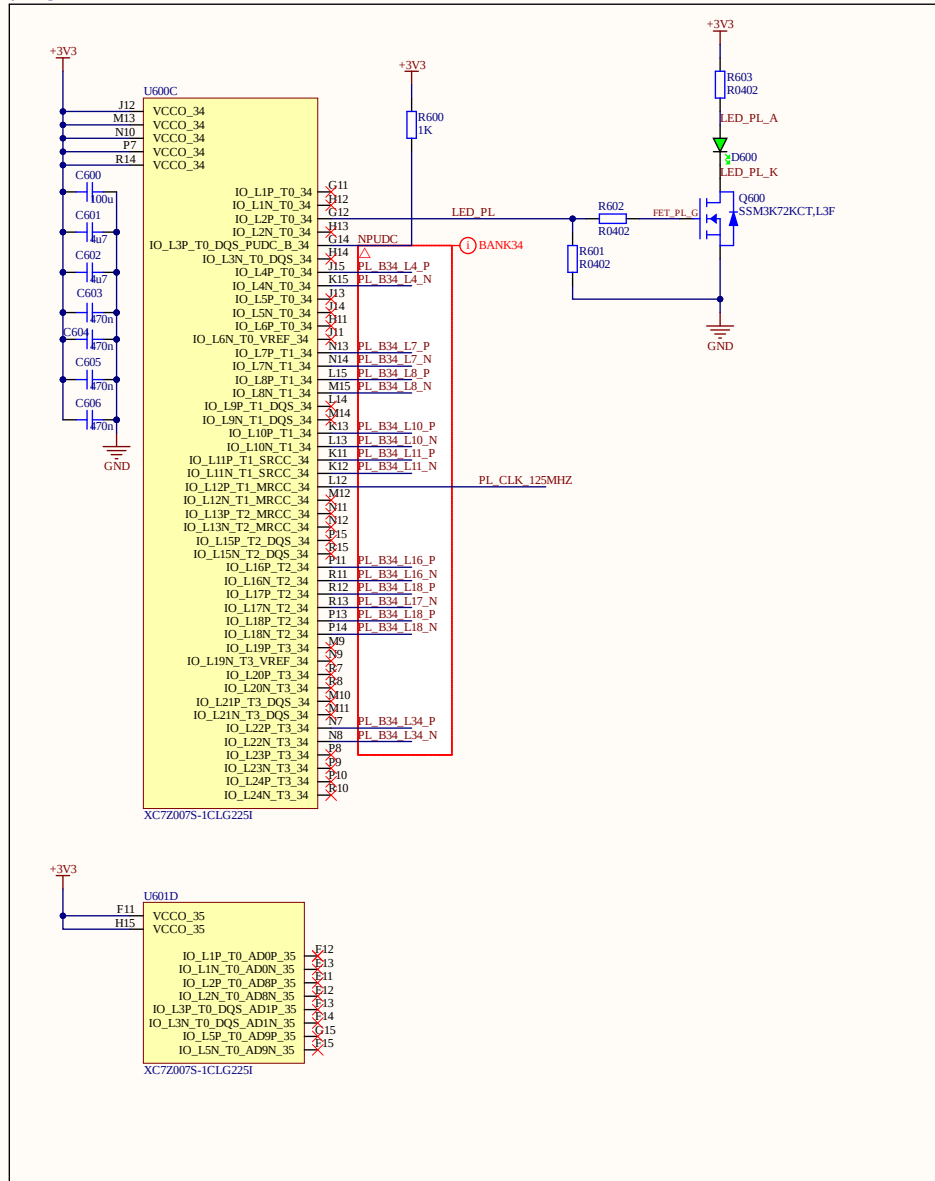
[5] Zynq DDR Interface & Termination

Zynq DDR Interface



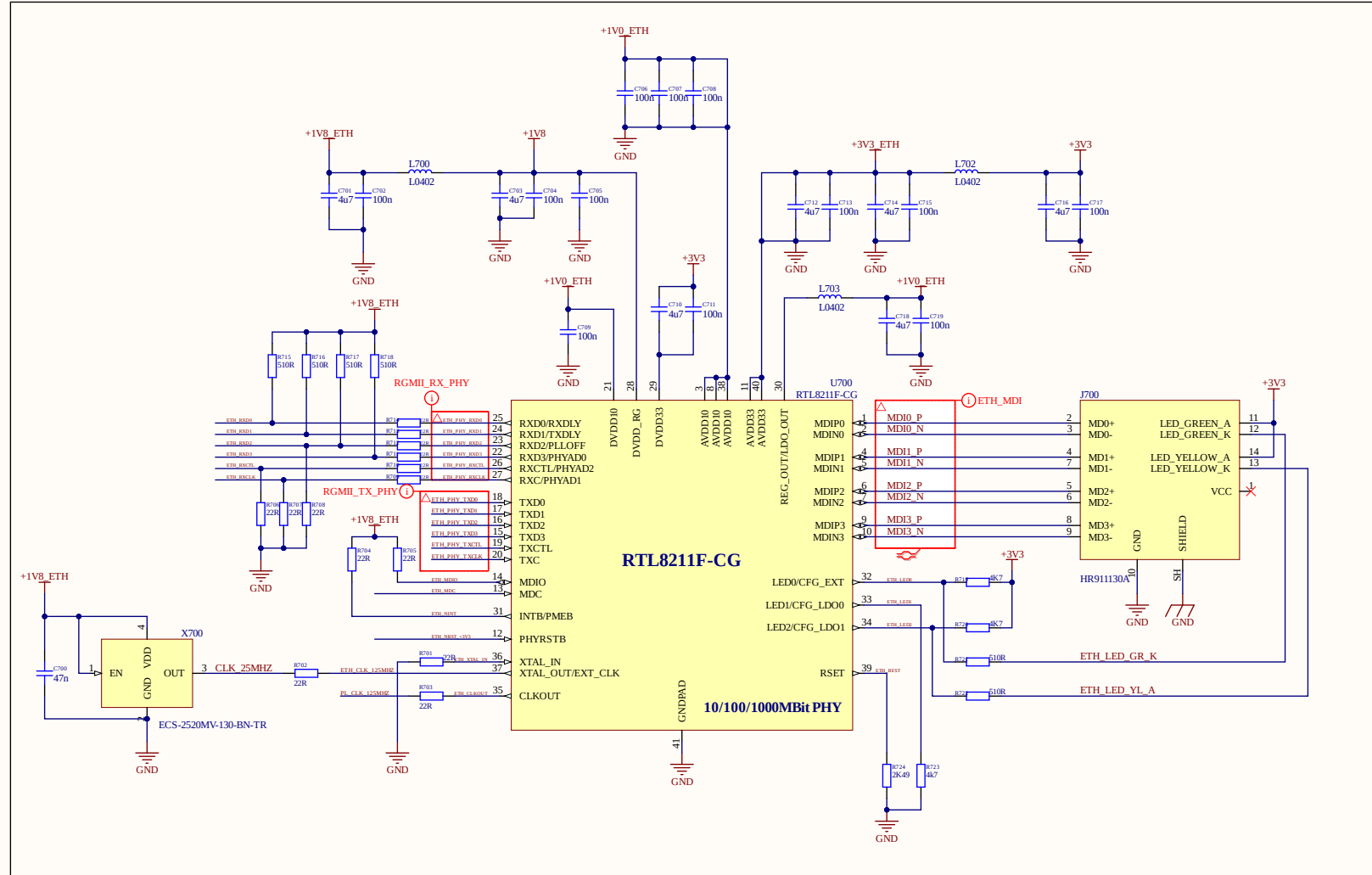
[6] Zynq Programmable Logic (PL)

Bank 34



[7] Gigabit Ethernet

Ethernet Phy & RJ45 Connector





[illegible]

[10] Mezzanine

